

AYR, Australia

WMO: 952950

Lat: 19.6169S

Lon: 147.3758E

Elev: 17

StdP: 101.12

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	10.2	0.2	3.8	18.8	2.9	4.7	17.6	11.8	25.2	8.3	22.0	2.0	200	0.340

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	33.1	24.7	32.2	24.6	31.5	24.3	27.5	30.0	26.6	29.8	26.0	29.2	4.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.5	28.4	25.7	21.1	28.1	25.2	20.3	27.7	87.2	30.3	83.1	29.9	80.6	29.3	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.4	5.5	35.8	1.4	1.5	4.5	36.9	3.7	37.8	2.9	38.6	1.9	39.7		
(5)			4.0	28.0	1.2	1.6	3.1	29.1	2.4	30.0	1.7	30.9	0.8	32.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	27.1	27.2	26.2	24.2	21.7	19.4	18.7	19.4	21.8	24.0	25.6	27.0	(6)
(7)		DBStd	3.51	1.26	1.34	1.18	1.28	1.99	2.38	2.22	1.93	1.65	1.62	1.37	1.35	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	54	0	0	0	0	3	17	23	11	1	0	0	0	(9)
(10)		CDD10.0	4931	531	481	503	427	362	281	271	292	354	434	469	528	(10)
(11)		CDD18.3	1944	272	247	244	177	107	48	35	45	104	176	219	269	(11)
(12)		CDH23.3	16492	2755	2459	2184	1234	571	212	159	288	730	1342	1866	2693	(12)
(13)		CDH26.7	5008	982	889	698	281	62	8	3	17	116	348	585	1021	(13)
(14)	Wind	WSAvg	3.0	3.1	3.1	3.1	3.1	2.8	2.8	2.8	3.0	3.3	3.4	3.2	(14)	
(15)	Precipitation	PrecAvg	1009	209	236	161	51	43	23	12	17	9	27	46	140	(15)
(16)		PrecMax	1754	954	651	543	430	191	212	56	115	63	237	215	505	(16)
(17)		PrecMin	366	15	3	2	0	0	0	0	0	0	0	0	11	(17)
(18)		PrecStd	392	201	170	137	83	51	35	16	25	17	44	54	138	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	35.0	32.6	30.9	29.2	27.8	27.0	28.2	30.6	32.5	32.9	33.9	(19)
(20)			MCWB	26.1	24.5	24.7	22.6	21.7	20.7	18.9	18.5	19.0	20.8	23.2	24.7	(20)
(21)		2%	DB	32.4	33.0	31.6	29.7	28.2	26.4	25.9	26.9	28.9	30.4	31.2	32.6	(21)
(22)			MCWB	25.3	25.1	24.6	22.7	21.2	19.9	18.3	18.6	20.2	22.4	23.1	24.6	(22)
(23)		5%	DB	31.6	31.9	30.8	29.1	27.3	25.6	25.1	26.0	27.9	29.4	30.5	31.8	(23)
(24)			MCWB	24.9	24.8	24.2	22.3	20.7	19.3	17.9	18.1	20.1	21.8	22.8	24.3	(24)
(25)		10%	DB	30.8	31.0	30.0	28.4	26.4	24.7	24.2	25.0	27.0	28.6	29.7	31.0	(25)
(26)			MCWB	24.6	24.6	23.9	22.1	20.2	18.5	17.6	17.6	19.6	21.2	22.5	24.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	27.8	28.6	27.9	24.8	24.5	21.7	22.1	23.4	24.5	26.2	29.8	(27)
(28)			MCDWB	30.0	30.4	29.4	28.2	27.4	25.5	24.0	25.6	27.1	28.8	29.2	30.0	(28)
(29)		2%	WB	27.0	26.7	26.5	25.4	23.2	21.7	20.4	20.9	22.4	23.8	25.1	27.1	(29)
(30)			MCDWB	30.6	30.1	29.2	26.6	26.3	24.5	23.8	24.4	26.1	27.8	28.6	30.3	(30)
(31)		5%	WB	26.2	26.2	25.6	24.1	22.1	20.6	19.6	19.9	21.5	23.1	24.4	26.1	(31)
(32)			MCDWB	29.6	29.5	28.8	26.8	25.6	23.9	23.3	23.9	25.7	27.3	28.1	29.5	(32)
(33)		10%	WB	25.7	25.8	24.9	23.2	21.1	19.6	18.8	19.0	20.7	22.5	23.7	25.2	(33)
(34)			MCDWB	28.9	28.9	28.2	26.6	24.9	23.2	22.6	23.2	25.4	26.8	27.6	28.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	7.4	7.9	8.8	10.0	10.8	11.7	12.0	11.2	9.6	8.4	8.0	(35)
(36)			MCDBR	8.5	9.2	9.0	9.6	10.5	10.8	11.7	12.5	11.9	10.2	9.1	8.9	(36)
(37)			MCWBR	3.1	3.3	3.5	4.1	4.9	5.2	5.3	5.4	4.9	3.5	2.8	2.9	(37)
(38)		5% WB	MCDBR	7.1	7.0	7.3	7.3	9.2	9.2	9.1	10.5	10.1	8.8	7.4	7.5	(38)
(39)			MCWBR	3.2	3.1	3.6	4.3	5.4	5.1	4.8	5.0	4.6	3.3	2.8	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.403	0.396	0.388	0.364	0.345	0.332	0.321	0.334	0.362	0.389	0.395	0.403	(40)	
(41)		taud	2.513	2.539	2.541	2.595	2.605	2.629	2.643	2.584	2.501	2.450	2.472	2.478	(41)	
(42)		Ebn at Noon	942	938	921	906	888	883	905	923	931	933	944	942	(42)	
(43)		Edh at Noon	114	110	106	93	86	81	82	93	108	119	118	118	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.41	6.00	5.57	5.02	4.30	3.88	4.36	5.18	6.17	6.77	6.91	6.89	(44)	
(45)		RadStd	0.53	0.63	0.46	0.43	0.29	0.34	0.28	0.31	0.34	0.38	0.73	0.62	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page